OPENNESS ON THE BASE OF THE LOCUS WHERE A BASE-CHANGE COMPATIBLE DIRECT IMAGE IS A FINITE LOCALLY FREE MODULE OF FIXED RANK IN A FIXED DEGREE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Reduction-type card: the span contains the complete argument; the two cited inputs (flat proper perfect direct image with base change, and the perfect openness locus) are stated as lemmas of the same chapter.

Editorial note. The label lemma-open-where-cohomology-in-degree-i-rank-r-geometric names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Cohomology and base change for a proper flat finitely presented morphism f : X to Y with F of finite presentation flat over Y: the object K = Rf_* F is perfect and its formation commutes with arbitrary base change (flat proper perfect direct image lemma)
- The perfect complex openness lemma of the same chapter: for a perfect object K of D(O_Y) and fixed i, r there is an open V of Y such that T to Y factors through V if and only if K restricted over T is isomorphic to a finite locally free module of rank r placed in degree i (open where cohomology in degree i rank r lemma)

Source reference. Original reference 0B91: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6442–6442. Prerequisite pointer invoked at 6780; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BDK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6743–6743. Prerequisite pointer invoked at 6784; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is proper, flat, and of finite presentation. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation, flat over Y. Fix $i, r \in \mathbf{Z}$. Then there exists an open subspace $V \subset Y$ with the following property: A morphism $T \to Y$ factors through V if and*

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only if $Rf_{T,*}\mathcal{F}_T$ is isomorphic to a finite locally free module of rank r placed in degree i.

Proof. By cohomology and base change (Lemma 0B91) the object $K = Rf_*\mathcal{F}$ is a perfect object of the derived category of Y whose formation commutes with arbitrary base change. Thus this lemma follows immediately from Lemma 0BDK. □